SmartCab Q-Learning Simulation

CMSC 436 • Project 3 • Pr. 3.2 – Question 1

Results

The following table summarizes the average rewards from 12 hyperparameter combinations. Each model was trained for 15,000 episodes and evaluated over 500 test episodes.

Run	Alpha	Gamma	Epsilon	Average reward
1	0.01	0.40	0.5	-94.31
2	0.01	0.40	1.0	-56.01
3	0.01	0.99	0.5	-22.79
4	0.01	0.99	1.0	-36.80
5	0.10	0.40	0.5	6.17
6	0.10	0.40	1.0	7.87
7	0.10	0.99	0.5	7.90
8	0.10	0.99	1.0	7.85
9	**0.90**	**0.40**	**0.5**	**8.03**
10	0.90	0.40	1.0	7.90
11	0.90	0.99	0.5	8.02
12	0.90	0.99	1.0	8.00

Best observed result: Run 9 — alpha = 0.90, gamma = 0.40, epsilon = 0.5; average reward = 8.03.

Discussion

For this part of the project, we tested different hyperparameters to see how they affected SmartCab's performance using Q-Learning. We changed alpha, gamma, and epsilon across 12 combinations and compared the average rewards after training for 15,000 episodes and testing for 500 episodes.

Changing the hyperparameters made a noticeable difference. When alpha was 0.01, all four combinations produced negative average rewards. The lowest was -94.31 at alpha = 0.01, gamma = 0.40, and epsilon = 0.5. This suggests the SmartCab learned too slowly to make consistently good decisions within the training period.

When we increased alpha to 0.1, all four average rewards became positive, ranging from 6.17 to 7.90. At alpha = 0.9, the rewards were consistently close to 8. The highest average reward was 8.03, using alpha = 0.9, gamma = 0.40, and epsilon = 0.5.

Gamma and epsilon also affected the results, although not in a consistent direction. For example, at alpha = 0.01 and epsilon = 0.5, increasing gamma from 0.40 to 0.99 improved the reward from -94.31 to -22.79. At alpha = 0.9, however, both gamma values produced very similar results.

Overall, alpha had the clearest impact in this set of experiments. Increasing the learning rate was associated with substantially better average rewards, while the effects of gamma and epsilon depended on the other settings.

What the Results Mean

The low learning rate of 0.01 performed poorly, while 0.9 consistently produced high rewards. All 12 experiments reported an average of zero illegal-action penalties during testing. However, the negative rewards in some runs suggest the taxi was taking too many steps or failing to complete trips within the step limit.

These results come from one set of randomized training runs. Because 8.03 and 8.02 are extremely close, we cannot say that the 8.03 combination will always perform better without repeating the experiments.

Conclusion

Our best observed combination was alpha = 0.9, gamma = 0.40, and epsilon = 0.5, with an average reward of 8.03. The experiment shows that hyperparameter choices can strongly influence how well SmartCab learns to complete trips.